ISYS 1108/ 3474 Software Engineering Project Management

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Senior Lecturer in Software Engineering

Week 6: Lectorial

What's next...

RMIT UNIVERSITY

Acknowledgement of Country

RMIT University acknowledges the people of the Woi wurrung and Boon wurrung language groups of the eastern Kulin Nation on whose unceded lands we conduct the business of the University.

RMIT University respectfully acknowledges their Ancestors and Elders, past and present.

RMIT also acknowledges the Traditional Custodians and their Ancestors of the lands and waters across Australia where we conduct our business.

Classes and Deadlines

Mid-semester break is between Weeks 5 and 6

	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12
Lectorials												
Workshops						Team-based assignment activities						
A1: SE Challenge				Friday: Challenge 1 10%					Friday: Challenge 2 10%		Friday: Challenge 3 10%	
A2: Team-based assignment								Due on Friday: 30%				
A3: Individual Assignment												Due on Friday: 40%

A2: Team-based Assignment

Assignments are team-based, unless you have corresponding ELP

Simulated Scrum Project: 30%

You team will be provided with scenarios, for which you will need to create Scrum artefacts (product backlog, sprint backlog, notes, etc.)

The teams will be formed by tutors during the workshop sessions.

Typical size of a team: 4-5 students enrolled in the same workshop class.

The core work will be done in Workshop sessions 6+7, under the observation of the tutor.

Please check the marking criteria (below you can see the first 2 criteria)

Criteria	Ratings				Pts
Trello/Jira board layout	**3 pts** **Full marks** The board has correct format/layout as expected for Trello/Jira approach, including the presentation on PBI IDs, using labels, etc.	**2 pts** **Good** There are some mistakes, but the solution is generally reasonable. See feedback from the marker.	**1 pts** **Needs improvement** The solution contains many critical mistakes. See feedback from the marker.	**0 pts** **No marks** No link to the board submitted, or the board is completely incorrect, or you haven't contributed to the creation of this artefact.	3 pts
Completeness of the product backlog	**3 pts** **Full Marks** The set of cards presenting all PBIs (all cards on the boards) covers all functionalities requested by the client, none of the PBIs is out of scope, all items are correctly sorted by priorities. The spreadsheet version of the product backlog matches the board.	**2 pts** **Good** There are some mistakes, but the solution is generally reasonable. See feedback from the marker.	**1 pts** **Needs improvement** The solution contains many critical mistakes or the spreadsheet-version of the product backlog is missing. See feedback from the marker.	**0 pts** **Incorrect** No link to the board submitted, or the product backlog is completely incorrect/missing, or you haven't contributed to the creation of the product backlog.	3 pts

Our plan for this session

- 😊 Lean: Detailed discussion
- 😊 Kanban methodology
- 😊 Assignments: Can we use Val or ChatGPT?
- 😊 Challenges of teamwork

+ any questions you have on related material ☺

LEAN

05.
PURSUIT
PERFECTION
04.
ESTABLISH
PULL
01.
DEFINE
VALUE
LEAN
PRINCIPLES
03.
CREATE
FLOW
02.
MAP VALUE
STREAM

History

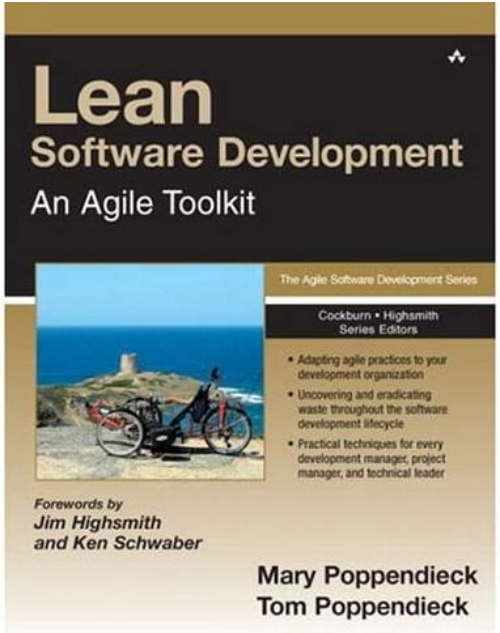
Lean was elaborated in Japan in the mid 1950-s in manufacturing industry (automotive industry - Toyota)

Lean was mainly aimed at

- waste reduction and
- sustainable production

In 2000-s Lean was also **adapted for software development** by Mary and Tom Poppendiecks

Lean was applied in the **startup industry** in 2008 by Eric Reis as a way of developing “new products and services in circumstances of extreme uncertainty.”

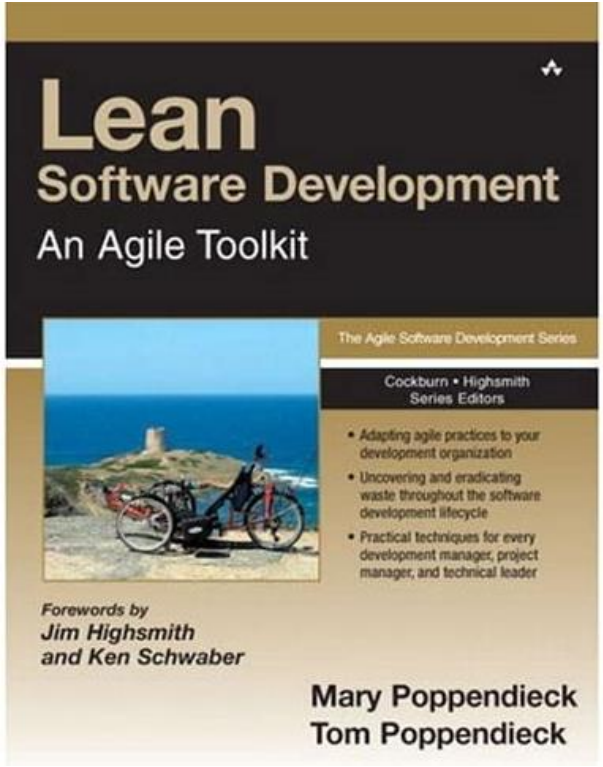


Understanding Lean

"Copying practices without understanding the underlying principles has a long history of mediocre results. But when the underlying principles are understood, it is useful to copy practices that work for similar organizations and modify them to fit your environment."

Mary and Tom Poppendieck,

Lean Software Development



5 Principles of Lean

1. **Define Value**
   - Understand *and* maximise value
   - Specify value as the customer sees it
2. **Map value stream**
   - Identify activities that contribute to the values
   - Activities that do not add value == waste
   - Eliminate waste
3. **Create flow**
   - Ensure that the flow of the remaining steps run smoothly without interruptions/ delays
4. **Establish pull**
   - limit inventory and work in process (WIP) items while ensuring that the requisite materials and information are available for a smooth flow of work
5. **Continuous improvement (*Kaizen*)**
   - Keep changing

1) Define value

Clearly identify objectives and requirements

Use them as acceptance criteria

Doing the right projects right == Satisfying the needs of all stakeholders.

"Make things as simple as possible, but not simpler."

–Albert Einstein

2) Map the value stream

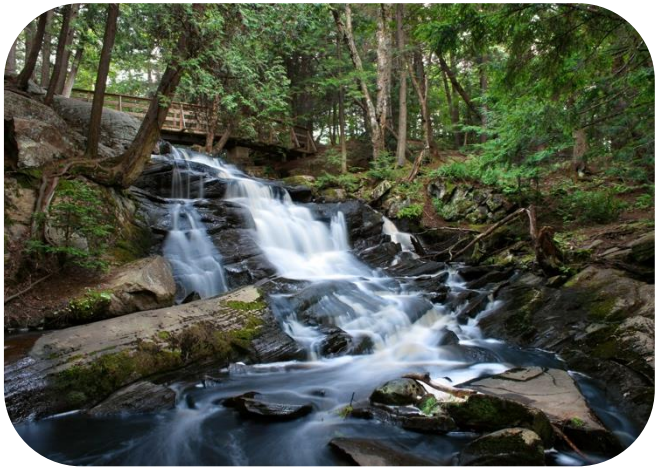
1. Identify value stream

- all the actions required to create a product,
- the process of “manufacturing” (NOT the product data flow or scenarios!)

2. Further identify

- Redundant steps
- Steps that burden resources
- Steps that impact risk, relationships and quality

3. Fine-tune the process



3) Create flow

Modelling can be involved:
- Process Modelling

The ability of a single order or piece of work to flow smoothly across the whole system without interruptions and at the maximum possible speed

The more things you think you are doing at the same time, the less productive you become (in realty, we are not as multi-tasking as we would like to be…)

- Gerald Weinberg "*Quality Software Management: Systems Thinking*"

Avoid committing teams to unrealistic schedules and budgets

Avoid unnecessary steps and interactions

4) Establish pull

Ensure that every project addresses a real meaningful need

Planning and executing cycles must be reduced to the minimum possible

Limit inventory and work in process (WIP) items for each stage

Similar idea we have in Scrum:

- Only completed items counted towards team's productivity
- Started but not completed (e.g., not tested properly) is seen as zero-contribution to the sprint

5) Kaizen

- Keep the strategic goals in focus
- Avoid short sighted “efficiencies”
- Measure performance
- Analyse results
- Improve the process
- Involve all stakeholders

改 **KAI=Change**

善 **ZEN=Good**

改善 **KAIZEN**

Continuous improvement

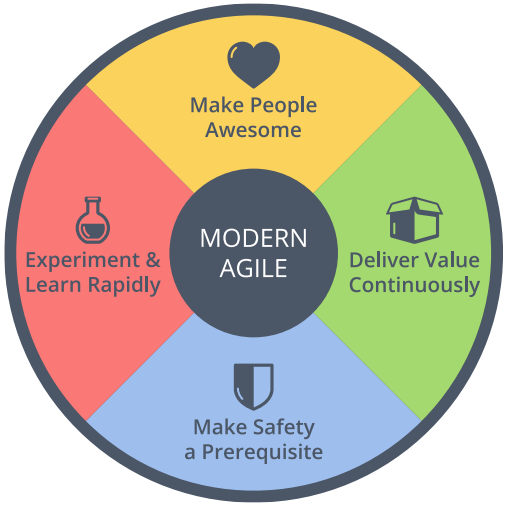
Lean skeptics will try to argue that “*perfection is impossible*” or even say “*why do we want to be better? We are just fine now.. ..*”

In a Lean environment, there is a perpetual state of discomfort with the current state: Lean leaders will constantly push the organisation out of its zone of comfort in search of a new, higher level of quality, performance, and, overall, client satisfaction.

Lean + Agile =

A methodology to deliver performance efficiency & effectiveness

- Free flowing, meaningful communication
- No excess and no insufficiency
- Self-managed teams
- Commitment to success
- Managed change & continuous improvement



Widely implemented and documented by many popular organizations, e.g., Microsoft's take on Lean: [https://msdn.microsoft.com/en-us/library/hh533841.aspx]

Agile Philosophy	Lean Philosophy
Is aimed at executing tasks faster, adapting to changes easier	Is about Smart development, when you improve virtually everything you do by eliminating anything that doesn't bring value to the customer
Makes the developing process flexible	Makes the developing process sustainable
Was initially designed for Software Development, then expanded to Marketing, and is currently applied in other areas	Started from traditional manufacturing and expanded to all existing industries
Action loop: product backlog – sprint backlog – iteration (sprints) – potentially shippable result	Action loop: build-measure-learn
Method for demonstrating progress — definition of 'done'	Method for demonstrating progress — validated learning
Methodologies: Scrum, XP, FDD, DSDM, Crystal Methods etc.	Methodologies: Kanban, Kaizen etc.
Toolkit: sprints, boards, Scrum Master, acceptance tests, user story mapping etc.	Toolkit: hypotheses, split (A/B) tests, customer interviews, funnel and cohort analysis, Customer Success Manager etc.

https://realtimeboard.com/blog/choose-between-agile-lean-scrum-kanban/

Implementation of Lean within Scrum

Visual workflow management

Sprint backlog as visual boards

Stand up meetings

Retro meetings (Kaizen!)

Risk mitigation

Quality control

Scrum	Kanban
fixed length sprints	a continuous flow model
defined roles	does not define any specific roles for the team
uses velocity as a key metric	uses of cycle time as a key metric

https://docs.microsoft.com/en-us/azure/devops/learn/agile/what-is-kanban

Visual tools for Lean project management

LeanKit

- [https://resources.leankit.com]

Trello

- [https://trello.com/]

Interesting article to read:

- [https://docs.microsoft.com/en-us/azure/devops/learn/agile/what-is-kanban]

Action time ☺

Recap: Agile and Lean

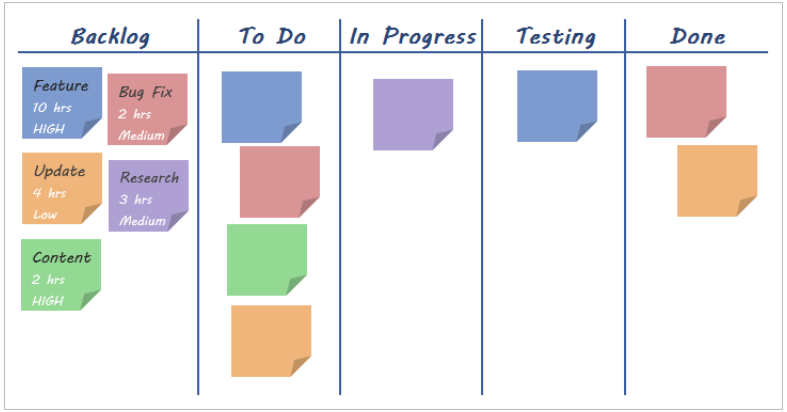
Due No due date **Points** 6 **Questions** 6
Time Limit None **Allowed Attempts** Unlimited

Instructions

This is ungraded, unlimited attempt quiz for you to ensure you have grasped the core concepts related to Agile and Lean.

Take the Quiz

Kanban methodology



History of Kanban

Derived from Japanese words ***Kan***, meaning visual, and ***Ban***, meaning card or board.

The person putting doors on cars only assembles enough doors to meet the demand of cars

When the inventory runs low, the person sends a demand signal (*Kanban card*) back to the previous operation that builds the doors

The person building the doors builds only enough doors to meet demand and in turn sends a demand signal for more materials to build the doors.

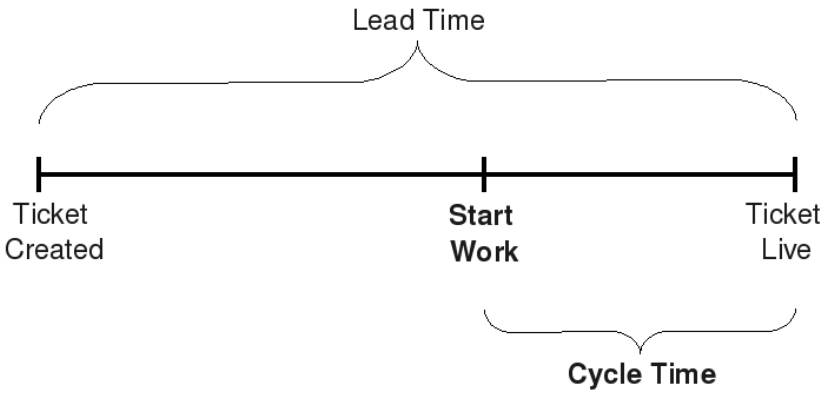
Kanban

It is used for demand-driven processes that are very reactive and difficult and/or impractical to plan:

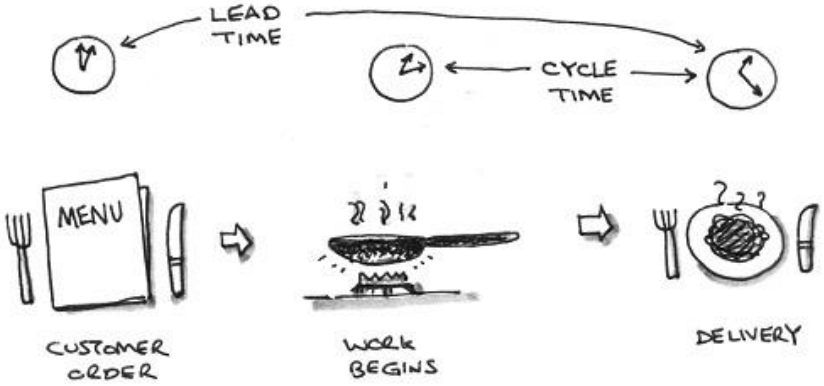
- A call centre queue for handling customer calls
- Some repetitive development processes that are heavily demand-driven, such as a support function that creates reports for business group
- An ongoing support process for providing continuous ongoing development of a product that include bug fixes and minor enhancements.

Kanban is sometimes used for software application development processes that need to be highly adaptive and may be difficult to plan in advance.

Kanban: Lead Time and Cycle Time



https://stefanroock.wordpress.com/2010/03/02/kanban-definition-of-lead-time-and-cycle-time/



Kanban in Scrum: Overlapping skills

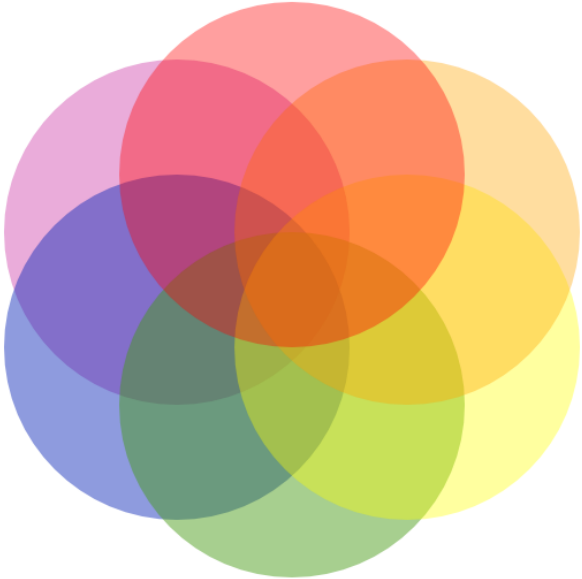
Overlapping skills = large benefit for the team

When only one person holds a skill set, that person becomes a bottleneck in the workflow.

Teams employ basic best practices like code review and mentoring help to spread knowledge.

Shared skills mean that

- team members can take on heterogeneous work, which further optimises cycle time
- if there is a backup of work, the entire team can swarm on it to get the process flowing smoothly again



Recap:
The roles in Scrum are PO, SM, and Team.
(not developers, testers, BA, …)

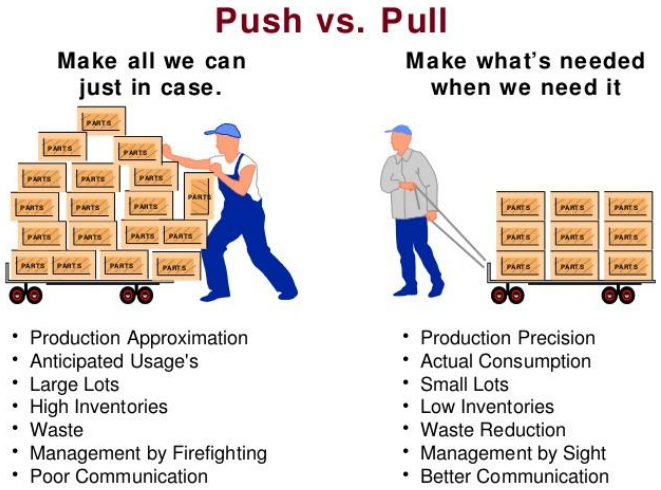
Push-system

A *push* system is totally planned in advance.

An example would be a traditional manufacturing process:

- It starts with a manufacturing plan based on a forecast of demand that someone thinks will take place.
- Raw materials are ordered and held in inventory waiting to go into production.
- Production resources are planned in advance and allocated to meet the forecasted demand in the manufacturing plan.

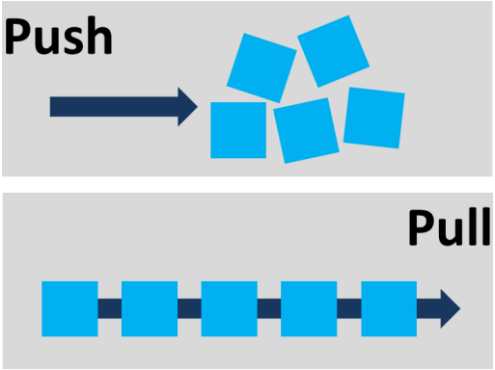
➔raw materials are planned and queued and pushed through the manufacturing process.



A traditional waterfall development process works on the same push principle: Requirements are forecast based on what someone thinks the customer needs, work is planned in advance, and resources are scheduled against the plan.

Pull-system

A *pull* system is heavily demand-driven and less-planned.



An example would be a manufacturing process where furniture is custom built to satisfy customer orders. In an ideal pull system:

- There may be a loosely defined plan to put capacity in place in anticipation of demand but no actual work is planned or scheduled until a customer order is received.
- Raw materials might be ordered only as needed to satisfy customer orders.
- Production resources would be allocated to work only as needed to satisfy customer orders.

➔ customer's orders are queued and raw materials are pulled through the process to meet actual demand

This can be seen as the inverse of a push process ☺

Kanban in Scrum: Pull-model, no multitasking

Once the team completes a work item, they pick the next work item off the top of the backlog.

In fact, savvy Product Owners always engage the Development Team when considering changes to the backlog.

Multitasking is discouraged as it kills efficiency.

The more work items in flight at any given time, the more context switching, which hinders their path to completion.

A Kanban team is only focused on the work that's actively in progress (WIP).

Key point of Kanban – **to limit the amount of *WIP***.

WIP limits highlight bottlenecks and backups in the team's process due to lack of focus, people, or skill sets.

Example: Kanban WIP

A typical software team might have four workflow states: to do, in progress, code review, and done.

They could choose to set a WIP limit of 2 for the code review state (i.e., they limit the number of items to be ”in progress” in the code review state to max. 2).

That might seem like a low limit, but there's good reason for it: code that hasn't been reviewed not only hasn't shipped yet, but may need significant re-work before it is ready to ship.

So it's important to take action on code reviews right away, and setting a WIP limit helps the team hold themselves accountable to that.

It forces the team to complete those reviews before pulling in new work.

Kanban	Scrum
Continues flow model, no sprints (Kanban board for Kanban → No column for Sprint Backlog)	Work in iterations: sprints have equal length to achieve cadence (Kanban board for Scrum → column for Sprint Backlog)
Items can be added to Product backlog anytime	Items can be added to Product backlog anytime
	Items can be added to Sprint backlog only at the Sprint Planning meeting
Estimation of work is not a part of Kanban process, but you can add it extra if you like.	Estimation of work is key element for planning – on its basis you analyse progress and decide scope of a sprint. Efforts are estimated in story points (Fib. numbers should be used).
Work in Progress (WIP) limits	WIP limits is not a part of Scrum process, Scrum implements this idea indirectly (PBI can be counted as completed only when it's done in terms of its DoD)
No prescribed roles: A Kanban process can be adapted to almost any kind of team structure	Roles: Scrum Master, Product Owner, Team
No prescribed format for requirements	Requirements should be specified in user story format

Action time ☺

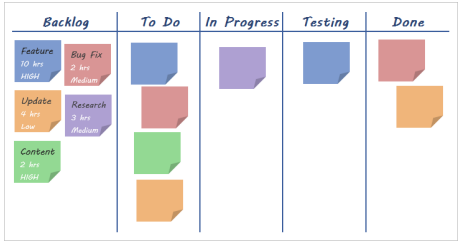
Recap: Kanban and Lean

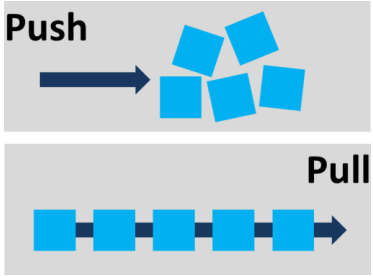
Due No due date **Points** 5 **Questions** 5
Time Limit None **Allowed Attempts** Unlimited

Instructions

This is ungraded, unlimited attempt quiz for you to ensure you have grasped the concepts related to Kanban and Lean.

Take the Quiz

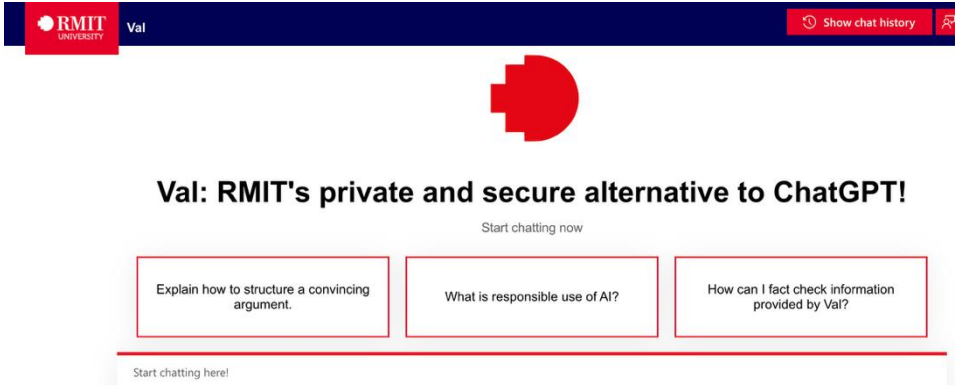




SEPM Assignments: Can we use Val, ChatGPT or other AI-based tools?

In short: yes, BUT

- You need to make sure that you provide corresponding note/reference in your submission to make clear that you used it **(otherwise it can be considered as an academic misconduct)**
- The responsibility for correctness of the solution is on you, not on the tool.

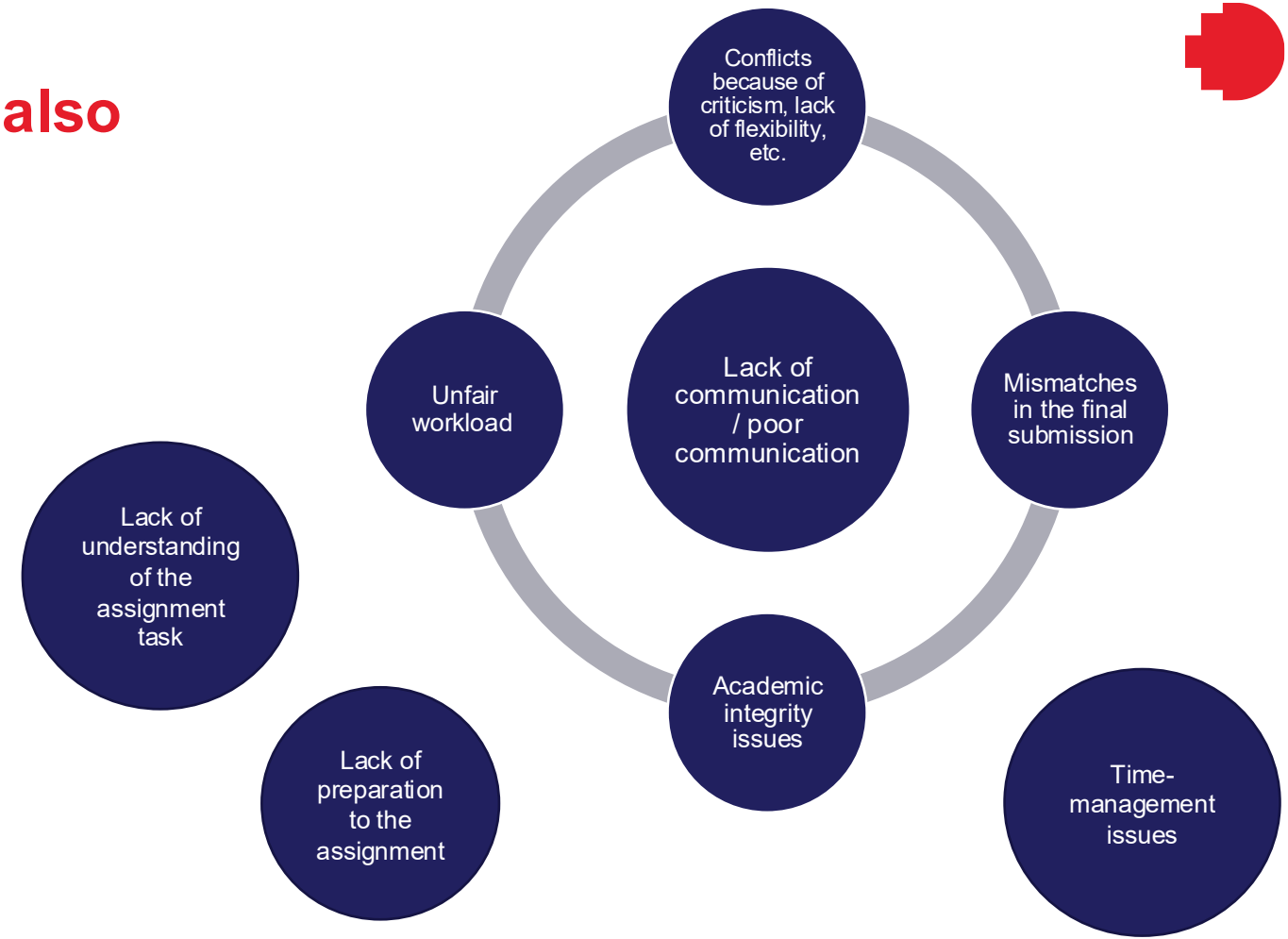


Let's now play with Val to see whether and how it might help us…

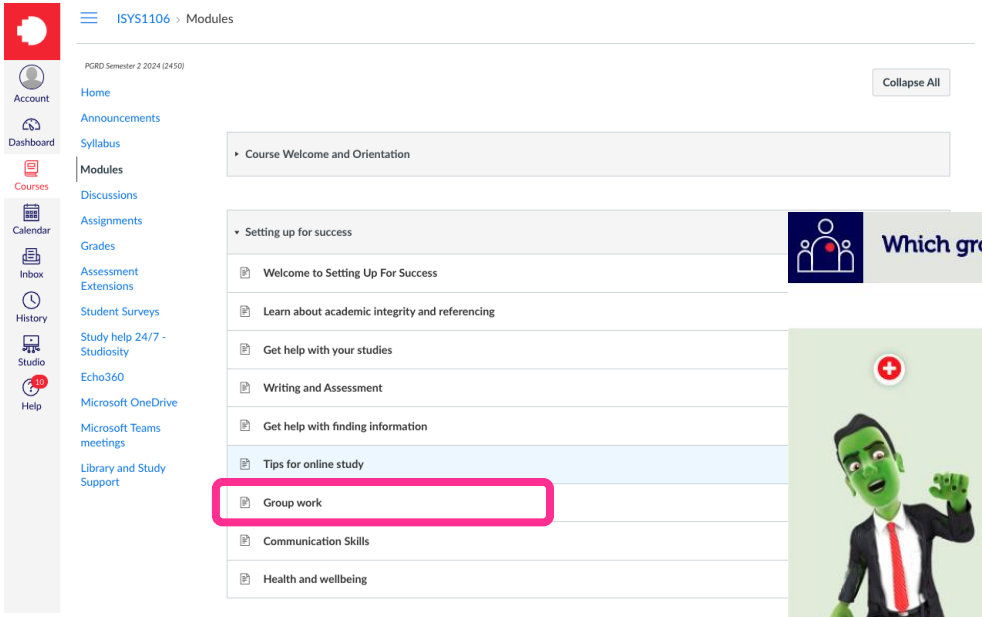
Challenges of teamwork (and how to overcome them)

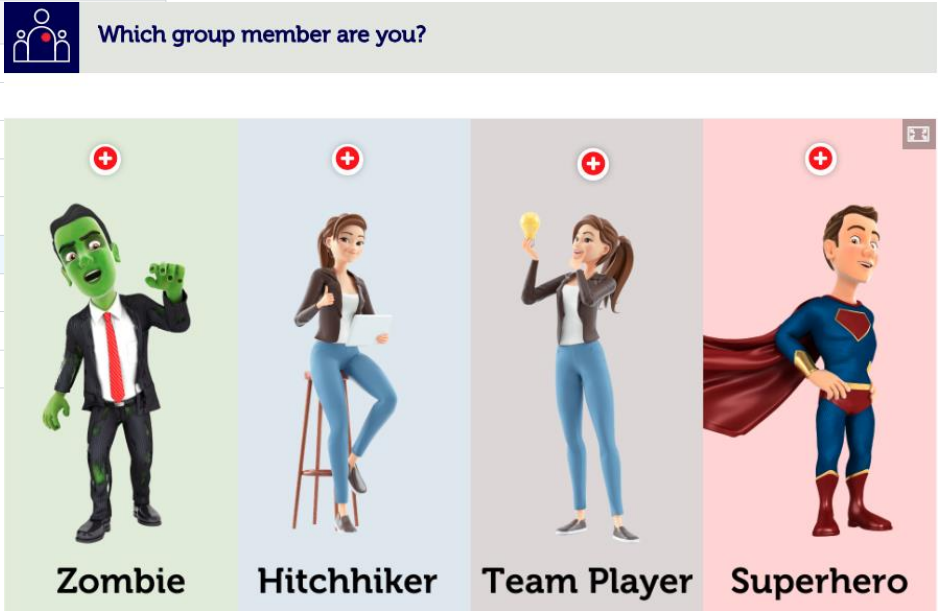
Let’s discuss!

Team work in assignments also might have challenges…



How to work effectively as an assignment team





Action time ☺

Let’s go to www.menti.com to discuss challenges in project team work and how to overcome them in traditional and Scrum projects ☺

Questions?

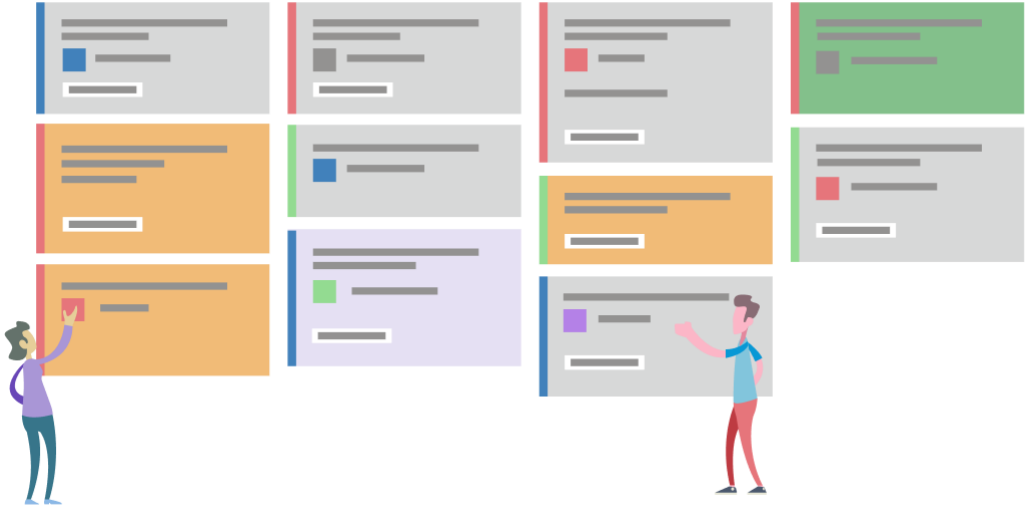


Figure from https://docs.microsoft.com/en-us/azure/devops/learn/agile/what-is-kanban